

## LONDON STANSTED, United Kingdom

WMO: 036830

Lat: 51.885N

Lon: 0.235E

Elev: 106

StdP: 100.06

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |            | 1%          |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-3.9 | (c)<br>-2.2 | (d)<br>-6.2                   | (e)<br>2.3 | (f)<br>-1.7 | (g)<br>-5.0 | (h)<br>2.5 | (i)<br>-0.8 | (j)<br>13.4           | (k)<br>9.1 | (l)<br>12.2 | (m)<br>9.0 | (n)<br>2.9            | (o)<br>350 | (p)<br>0.608 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 27.2            | 18.5 | 25.1 | 17.7 | 23.2 | 16.8 | 19.5                | 25.1 | 18.6 | 23.5 | 17.7 | 21.9 | 4.4                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.8                            | 12.9 | 21.5 | 16.9 | 12.2 | 20.5 | 16.0 | 11.5 | 19.5 | 56.1          | 25.8 | 52.9 | 23.6 | 50.2 | 22.1 | 24.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 11.1              | 9.7  | 8.6 | -6.6                       | 31.0 | 2.5                | 1.9 | -8.5                                               | 32.3 | -9.9       | 33.4 | -11.3      | 34.5 | -13.2      | 35.9 |
|                   |      |     | -7.0                       | 21.5 | 2.5                | 1.1 | -8.8                                               | 22.3 | -10.2      | 22.9 | -11.6      | 23.5 | -13.4      | 24.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.2        | 4.1   | 4.5   | 6.4   | 8.8   | 11.8  | 14.9  | 17.3  | 17.1  | 14.4  | 11.0  | 7.1   | 4.7   |
|                                                                        | DBStd       | 5.60        | 3.21  | 3.27  | 3.05  | 3.11  | 3.06  | 2.75  | 2.73  | 2.66  | 2.57  | 2.96  | 3.12  | 3.60  |
|                                                                        | HDD10.0     | 816         | 182   | 154   | 117   | 59    | 16    | 1     | 0     | 0     | 1     | 23    | 96    | 168   |
|                                                                        | HDD18.3     | 3015        | 440   | 386   | 371   | 287   | 203   | 109   | 53    | 57    | 121   | 227   | 337   | 424   |
|                                                                        | CDD10.0     | 892         | 1     | 1     | 4     | 22    | 72    | 147   | 227   | 220   | 133   | 55    | 8     | 3     |
|                                                                        | CDD18.3     | 49          | 0     | 0     | 0     | 0     | 1     | 5     | 22    | 18    | 3     | 1     | 0     | 0     |
|                                                                        | CDH23.3     | 424         | 0     | 0     | 0     | 1     | 6     | 56    | 192   | 147   | 19    | 2     | 0     | 0     |
|                                                                        | CDH26.7     | 89          | 0     | 0     | 0     | 0     | 1     | 9     | 42    | 36    | 2     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.3         | 4.9   | 4.9   | 4.7   | 4.2   | 4.2   | 3.9   | 3.8   | 3.8   | 3.9   | 4.3   | 4.3   | 4.6   |
| (15) Precipitation                                                     | PrecAvg     | 623         | 55    | 40    | 46    | 45    | 45    | 48    | 52    | 54    | 58    | 59    | 60    | 59    |
|                                                                        | PrecMax     | 787         | 126   | 82    | 99    | 106   | 87    | 126   | 110   | 105   | 131   | 154   | 170   | 120   |
|                                                                        | PrecMin     | 459         | 10    | 12    | 4     | 8     | 6     | 3     | 4     | 3     | 4     | 3     | 20    | 16    |
|                                                                        | PrecStd     | 89          | 27    | 21    | 25    | 22    | 23    | 30    | 25    | 28    | 37    | 41    | 30    | 28    |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.2  | 14.0  | 16.8  | 21.9  | 24.1  | 27.9  | 30.2  | 30.0  | 26.0  | 21.0  | 15.8  | 13.8  |
|                                                                        |             | MCWB        | 10.8  | 11.1  | 11.3  | 14.7  | 16.8  | 18.6  | 19.5  | 19.4  | 17.6  | 16.8  | 14.2  | 11.9  |
|                                                                        | 2%          | DB          | 11.2  | 11.9  | 14.2  | 18.2  | 22.0  | 24.9  | 27.6  | 27.1  | 22.8  | 18.0  | 14.1  | 12.2  |
|                                                                        |             | MCWB        | 9.9   | 9.9   | 10.4  | 12.5  | 15.7  | 17.6  | 18.6  | 18.5  | 16.8  | 15.0  | 12.7  | 10.9  |
|                                                                        | 5%          | DB          | 10.2  | 10.8  | 12.8  | 16.1  | 19.9  | 22.5  | 25.2  | 24.2  | 20.8  | 16.8  | 13.0  | 11.5  |
|                                                                        |             | MCWB        | 9.1   | 9.0   | 9.8   | 11.5  | 14.6  | 16.4  | 17.7  | 17.5  | 16.0  | 14.4  | 11.6  | 10.2  |
|                                                                        | 10%         | DB          | 9.1   | 9.3   | 11.2  | 14.2  | 17.9  | 20.8  | 23.2  | 22.2  | 19.0  | 15.4  | 11.9  | 10.1  |
|                                                                        |             | MCWB        | 8.2   | 8.0   | 8.7   | 10.3  | 13.3  | 15.6  | 17.0  | 16.4  | 14.8  | 13.5  | 10.6  | 9.1   |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 11.6  | 11.6  | 12.4  | 15.1  | 18.3  | 20.2  | 21.1  | 20.7  | 19.4  | 17.1  | 14.5  | 12.5  |
|                                                                        |             | MCDB        | 12.3  | 13.5  | 15.2  | 20.7  | 22.5  | 26.2  | 28.1  | 27.9  | 23.6  | 20.3  | 15.5  | 13.5  |
|                                                                        | 2%          | WB          | 10.5  | 10.2  | 11.2  | 13.3  | 16.3  | 18.3  | 19.5  | 19.3  | 17.8  | 15.7  | 13.1  | 11.5  |
|                                                                        |             | MCDB        | 11.3  | 11.3  | 13.5  | 17.3  | 20.6  | 22.8  | 25.3  | 25.0  | 21.3  | 17.5  | 14.1  | 12.5  |
|                                                                        | 5%          | WB          | 9.3   | 9.2   | 10.3  | 12.1  | 15.0  | 17.1  | 18.6  | 18.3  | 16.7  | 14.9  | 11.9  | 10.5  |
|                                                                        |             | MCDB        | 10.0  | 10.2  | 12.1  | 15.5  | 19.1  | 21.4  | 23.9  | 23.1  | 19.4  | 16.3  | 13.0  | 11.5  |
|                                                                        | 10%         | WB          | 8.2   | 8.2   | 9.3   | 10.9  | 13.9  | 16.1  | 17.6  | 17.4  | 15.7  | 14.0  | 10.8  | 9.3   |
|                                                                        |             | MCDB        | 9.0   | 9.3   | 11.0  | 13.7  | 17.1  | 19.8  | 22.2  | 21.4  | 18.2  | 15.3  | 11.8  | 10.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.2   | 5.8   | 7.2   | 8.8   | 9.1   | 9.3   | 9.8   | 9.4   | 8.5   | 7.0   | 5.7   | 4.9   |
|                                                                        |             | MCDBR       | 6.0   | 6.8   | 9.3   | 11.6  | 12.4  | 12.6  | 13.1  | 12.3  | 10.6  | 7.6   | 5.9   | 5.2   |
|                                                                        | 5% WB       | MCWBR       | 5.8   | 5.4   | 6.1   | 6.7   | 7.1   | 6.5   | 5.6   | 5.3   | 5.5   | 5.2   | 5.2   | 5.0   |
|                                                                        |             | MCWBR       | 6.0   | 6.2   | 8.2   | 10.5  | 11.2  | 11.1  | 11.6  | 10.7  | 8.9   | 6.8   | 5.7   | 5.3   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.324 | 0.340 | 0.375 | 0.397 | 0.402 | 0.405 | 0.402 | 0.392 | 0.372 | 0.356 | 0.328 | 0.314 |
|                                                                        |             | taud        | 2.377 | 2.387 | 2.282 | 2.238 | 2.261 | 2.278 | 2.295 | 2.346 | 2.400 | 2.433 | 2.425 | 2.364 |
|                                                                        | Ebn at Noon |             | 676   | 776   | 811   | 837   | 851   | 850   | 845   | 835   | 809   | 747   | 668   | 620   |
|                                                                        |             | Edh at Noon | 61    | 80    | 106   | 124   | 127   | 127   | 123   | 111   | 94    | 75    | 58    | 53    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.77  | 1.44  | 2.55  | 4.02  | 4.84  | 5.28  | 5.03  | 4.20  | 3.15  | 1.82  | 0.94  | 0.60  |
|                                                                        | RadStd      |             | 0.07  | 0.21  | 0.31  | 0.41  | 0.34  | 0.44  | 0.49  | 0.32  | 0.26  | 0.17  | 0.10  | 0.08  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | +0.41       | N/A     | N/A     | N/A     |
| (47) Regional (82 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page